Monotonicity: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Monotonic function and monotonicity at a point

Formulas

Increasing at a: $f(a-h) < f(a) < f(a+h)$; decreasing at a: $f(a-h) > f(a) > f(a+h)$

KEY POINTS

- Monotonic = only increasing or only decreasing in the whole domain
- Sufficient: $f'(a) > 0 \Rightarrow$ increasing at a; $f'(a) < 0 \Rightarrow$ decreasing at a; $f'(a) = 0$: look at the sign of f' on both sides
- End point of the domain: compare only with the side inside the domain
- Defined only in the domain; continuity / differentiability not needed

2. Monotonicity in an interval

Formulas

$x_1 > x_2 \Rightarrow f(x_1) > f(x_2)$ (increasing), $f(x_1) \geq f(x_2)$ (non-decreasing)
$x_1 > x_2 \Rightarrow f(x_1) < f(x_2)$ (decreasing), $f(x_1) \leq f(x_2)$ (non-increasing)

KEY POINTS

- IIT 'increasing' = strictly increasing; NCERT 'increasing' = non-decreasing
- Increasing on I_1 and on I_2 does not mean increasing on $I_1 \cup I_2$ $(-\frac{1}{x})$

3. Derivative test and intervals of monotonicity

Formulas

$f'(x) \geq 0$ on I (zero only at isolated points) $\Rightarrow f$ increasing; $f'(x) \leq 0 \Rightarrow f$ decreasing

STEPS

1. Find the domain
2. Factorise $f'(x)$; mark its zeros, the points where it does not exist and the gaps of the domain
3. Sign scheme of f' (even powers do not change sign)
4. Include a boundary point if it is in the domain and f is continuous there

KEY POINTS

- Piecewise: each piece separately, then compare $f(a^-)$, $f(a)$, $f(a^+)$ at a junction
- $f(g(x))$: same nature $\Rightarrow$ increasing, opposite nature $\Rightarrow$ decreasing
- f and f^{-1} have the same monotonic nature; a strictly monotonic continuous f is one-one

Watch out

Do not divide by an expression whose sign is unknown; points outside the domain split intervals

4. Critical and stationary points

KEY POINTS

- Critical point: in the domain, $f' = 0$ or f' does not exist
- Stationary point: $f' = 0$ (horizontal tangent); stationary $\subset$ critical
- $|x|$: 0; $x^{2/3}$: 0; $\{x\}$: integers; $[x]$, $\operatorname{sgn} x$: every real; $\tan x$, e^x: none
- Points outside the domain are never critical

5. Always increasing / decreasing (parameter questions)

Formulas

Cubic $ax^3 + bx^2 + cx + d$ monotonic on $\mathbb{R} \iff b^2 - 3ac \leq 0$

STEPS

1. Write $f'(x) \geq 0$ (or ≤ 0) for all x of the interval

2. Separate the parameter: $p \geq \max g(x)$ or $p \leq \min g(x)$

3. Or treat f' as a quadratic in x, $\sin x$, e^x... and use $a > 0$, $D \leq 0$ / location of roots

4. Check the boundary value of the parameter separately ($f' = 0$ on an interval is not allowed)

Watch out

'No critical points' needs $f' \neq 0$ everywhere: strict inequality
Coefficient of x^2 in f' can be zero: test that case separately

6. Greatest and least value; comparison of numbers

Formulas

$\left(\frac{\ln x}{x}\right)' = \frac{1 - \ln x}{x^2}$: $\uparrow$ on $(0, e]$, $\downarrow$ on $[e, \infty)$; $e \leq a < b \Rightarrow a^b > b^a$

KEY POINTS

- Continuous f on $[a, b]$: greatest and least values occur at critical points or at a, b
- Monotonic on $[a, b]$: the values are $f(a)$ and $f(b)$; range $= [f(a), f(b)]$ (or reversed)
- Range of a cubic on an interval: values at the end points and at the roots of $f' = 0$ inside it
- Greatest term of a sequence $a_n = f(n)$: study $f(x)$, then check the integers next to the turning point

Watch out

On an open interval the greatest / least value may not exist

7. Proving inequalities and solving $f(A) > f(B)$

STEPS

1. Put $h(x) =$ LHS $-$ RHS
2. Find h at the starting point ($h(0)$ or $h(a)$)
3. Show $h' > 0$ (or < 0) after that point; if unclear, repeat with h''

KEY POINTS

- Standard: $x > \sin x$, $\tan x > x$ $\left(0 < x < \frac{\pi}{2}\right)$, $\ln(1 + x) < x$, $e^x > 1 + x$ $(x \neq 0)$
- f increasing: $f(A) > f(B) \iff A > B$; f decreasing: $f(A) > f(B) \iff A < B$ (with A, B in the domain)

8. Rolle's theorem

Formulas

f continuous on $[a, b]$, differentiable on (a, b), $f(a) = f(b) \Rightarrow f'(c) = 0$ for some $c \in (a, b)$

KEY POINTS

- All three conditions together; they are sufficient, not necessary
- c is never an end point; at least one c, possibly more
- Geometry: a horizontal tangent between two equal ordinates
- Between two consecutive roots of $f = 0$ lies a root of $f' = 0$
- $f = 0$ has n roots $\Rightarrow f' = 0$ at least $n - 1$; $f' = 0$ has m roots $\Rightarrow f = 0$ at most $m + 1$

STEPS

1. Read the target as $g'(c) = 0$
2. $f' + kf \to e^{kx} f$; $cf'(c) - f(c) \to \frac{f}{x}$; $f' = k\phi' \to f - k\phi$; $(ff')'$, $(f'g')'$ for $f'^2 + ff''$, $f''g' + f'g''$
3. Polynomial condition on coefficients: integrate the polynomial and check equal values at two points

Watch out

Check differentiability inside: $|x|$, $[x]$, $x^{2/3}$ break it

9. Lagrange's mean value theorem (LMVT)

Formulas

$$f'(c) = \frac{f(b) - f(a)}{b - a},\ c \in (a, b); \qquad f(a + h) = f(a) + h\, f'(a + \theta h),\ 0 < \theta < 1$$

$$m \le f' \le M \Rightarrow m(b - a) \le f(b) - f(a) \le M(b - a)$$

KEY POINTS

- Conditions: continuous on $[a, b]$, differentiable on (a, b)
- Proof: Rolle on $g = f - \frac{f(b)-f(a)}{b-a}x$
- Geometry: tangent parallel to the chord AB; physics: instantaneous rate = average rate
- Quadratic: $c = \frac{a+b}{2}$
- Line meets the graph at m points $\Rightarrow f'' = 0$ at least $m - 2$ times (LMVT, then Rolle on f')

Watch out

Inequality via LMVT: identify $\frac{f(b)-f(a)}{b-a}$, then bound $f'(c)$ by the monotonicity of f'

10. Number of real roots

STEPS

1. Find the critical points; on each interval between them f is monotonic
2. Write f at the critical points and the limits at the ends
3. Count the sign changes: one root per monotonic piece with a sign change

KEY POINTS

- $f' > 0$ everywhere and $f \to \mp\infty$: exactly one real root (odd-degree with positive coefficients: $x^7 + 5x^3 + 3x + 1$)
- e^x-polynomials: put $t = e^x > 0$ and count only positive roots
- Parameter k in $f(x) = k$: count the intersections of $y = k$ with the graph

11. Plotting graphs

STEPS

1. Domain and symmetry
2. Limits at the boundary of the domain and at $\pm\infty$
3. Intervals of monotonicity and values at critical points
4. Intercepts and asymptotes; then sketch

KEY POINTS

- Cubic: $b^2 - 3ac \le 0$ monotonic; > 0 one local max and one local min
- Q/Q: vertical asymptotes from the denominator, $y = \frac{a_1}{a_2}$, turning values give the range
- $x^{1/x}$: greatest value $e^{1/e}$ at $x = e$; x^x: least value $e^{-1/e}$ at $x = \frac{1}{e}$

12. Asymptotes

Formulas

Vertical $x = a$: $f \to \pm\infty$ as $x \to a^{\pm}$; horizontal $y = a$: $\lim\limits_{x \to \pm\infty} f = a$

Inclined $y = mx + b$: $m = \lim \dfrac{f(x)}{x}$, $b = \lim(f(x) - mx)$

KEY POINTS

- Rational $\frac{P}{Q}$: vertical asymptotes at the real roots of Q (after cancelling)
- $\deg P < \deg Q$: $y = 0$; equal degrees: $y =$ ratio of leading coefficients
- $\deg P = \deg Q + 1$: divide, the quotient $ax + b$ is the inclined asymptote

13. Common mistakes - checklist

STEPS

1. Domain first: monotonicity, critical points and asymptotes live only in the domain
2. $f' = 0$ at isolated points does not stop f from being increasing
3. Union of intervals: check the junction before writing $\cup$
4. Rolle / LMVT: check every condition and keep c strictly inside
5. 'Increasing' in the question: decide strict / non-strict from the options